高等量子力学　第 2 次作业

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题目六　自旋为 1/2 的粒子，其三个自旋分量算符在 s_z 表象下的矩阵表示分别为

$$\hat{s}_x = \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \hat{s}_y = \frac{\hbar}{2}\begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \qquad \hat{s}_z = \frac{\hbar}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{1}$$

根据幺正变换矩阵的定义式：

(a) 求由 s_z 表象到 s_y 表象的幺正变换矩阵 S_{zy}；

(b) 求由 s_z 表象到 s_x 表象的幺正变换矩阵 S_{zx}。

回答

(a) 对于两个力学量算符 $\hat{A}, \hat{B}$，设它们的本征方程分别为

$$\hat{A}\left|a_i\right\rangle = a_i\left|a_i\right\rangle, \qquad \hat{B}\left|b_i\right\rangle = b_i\left|b_i\right\rangle, \tag{2}$$

则从 A 表象到 B 表象的幺正变换算符 $\hat{S}_{A\to B}$ 满足

$$\hat{S}_{A\to B} = \sum_i \left|b_i\right\rangle\left\langle a_i\right|, \qquad \hat{S}_{A\to B}\left|a_i\right\rangle = \left|b_i\right\rangle, \qquad \left(S_{A\to B}\right)_{ij} = \left\langle a_i \middle| b_j\right\rangle. \tag{3}$$

因此，s_z 表象到 s_y 表象的幺正变换矩阵 $S_{z\to y}$ 形式上可写为

$$S_{z\to y} = \begin{pmatrix} \langle +_z|+_y\rangle & \langle +_z|-_y\rangle \\ \langle -_z|+_y\rangle & \langle -_z|-_y\rangle \end{pmatrix}. \tag{4}$$

s_z 表象下，$\hat{s}_z, \hat{s}_y$ 本征态的矩阵表示为

$$|+_z\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |-_z\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \qquad |+_y\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ \mathrm{i} \end{pmatrix}, \qquad |-_y\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} -1 \\ \mathrm{i} \end{pmatrix}. \tag{5}$$

内积为

$$\langle +_z|+_y\rangle = \frac{1}{\sqrt{2}}, \qquad \langle +_z|-_y\rangle = -\frac{1}{\sqrt{2}}, \qquad \langle -_z|+_y\rangle = \frac{\mathrm{i}}{\sqrt{2}}, \qquad \langle -_z|-_y\rangle = \frac{\mathrm{i}}{\sqrt{2}}, \tag{6}$$

于是 $S_{z\to y}$ 为

$$S_{z\to y} = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{\mathrm{i}}{\sqrt{2}} & \frac{\mathrm{i}}{\sqrt{2}} \end{pmatrix}. \tag{7}$$

(b) s_z 表象下，$\hat{s}_x$ 本征态的矩阵表示为

$$|+_x\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad |-_x\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}. \tag{8}$$

于是由 s_z 表象到 s_x 表象的幺正变换矩阵 $S_{z\to x}$ 为

$$S_{z\to x} = \begin{pmatrix} \langle +_z|+_x\rangle & \langle +_z|-_x\rangle \\ \langle -_z|+_x\rangle & \langle -_z|-_x\rangle \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{pmatrix}. \tag{9}$$

题目七 从 $\hat{a}\,|0\rangle = 0$ 出发，求谐振子动量波函数中基态 Ψ_{p0} 及第一激发态 Ψ_{p1} 的具体表达式，其中 $\Psi_{pn} = \langle p\,|\,n\rangle$。

回答 湮灭算符可用坐标、动量算符表达：

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} + \frac{\mathrm{i}}{m\omega}\hat{p}\right). \tag{10}$$

$\hat{a}\,|0\rangle = 0$ 左乘 $\langle p|$ 有：

$$\langle p\,|\,\hat{x}\,|\,0\rangle + \frac{\mathrm{i}}{m\omega}\langle p\,|\,\hat{p}\,|\,0\rangle = 0. \tag{11}$$

注意到：

$$\langle p\,|\,\hat{p}\,|\,0\rangle = p\,\langle p\,|\,0\rangle = p\Psi_{p0}, \tag{12}$$

$$\langle p\,|\,\hat{x}\,|\,0\rangle = \int \mathrm{d}p'\,\langle p\,|\,\hat{x}\,|\,p'\rangle\,\langle p'\,|\,0\rangle = \int \mathrm{d}p'\,\langle p\,|\,\hat{x}\,|\,p'\rangle\,\Psi_{p'0}, \tag{13}$$

其中

$$\begin{aligned}
\langle p\,|\,\hat{x}\,|\,p'\rangle &= \int \mathrm{d}x\mathrm{d}x'\,\langle p\,|\,x\rangle\,\langle x\,|\,\hat{x}\,|\,x'\rangle\,\langle x'\,|\,p'\rangle = \frac{1}{2\pi\hbar}\int \mathrm{d}x\mathrm{d}x' x\mathrm{e}^{\mathrm{i}(p'x'-px)/\hbar}\delta(x-x') \\
&= \frac{1}{2\pi\hbar}\int \mathrm{d}x x\mathrm{e}^{\mathrm{i}(p'-p)x/\hbar} = \frac{1}{2\pi\hbar}\frac{\hbar}{\mathrm{i}}\int \mathrm{d}x\frac{\partial}{\partial\,(p'-p)}\mathrm{e}^{\mathrm{i}(p'-p)x/\hbar} \\
&= \frac{1}{2\pi\mathrm{i}}\frac{\mathrm{d}}{\mathrm{d}\,(p'-p)}\int \mathrm{d}x\mathrm{e}^{\mathrm{i}(p'-p)x/\hbar} = \frac{1}{\mathrm{i}}\frac{\mathrm{d}}{\mathrm{d}\,(p'-p)}\delta\left(\frac{p'-p}{\hbar}\right) \\
&= -\mathrm{i}\hbar\frac{\mathrm{d}\delta(p'-p)}{\mathrm{d}\,(p'-p)} = -\mathrm{i}\hbar\delta'(p'-p).
\end{aligned} \tag{14}$$

于是

$$\begin{aligned}
\langle p\,|\,\hat{x}\,|\,0\rangle &= \int \mathrm{d}p'\,\langle p\,|\,\hat{x}\,|\,p'\rangle\,\Psi_{p'0} = -\mathrm{i}\hbar\int \mathrm{d}p'\delta'(p'-p)\Psi_{p'0} \\
&= \mathrm{i}\hbar\frac{\mathrm{d}}{\mathrm{d}p}\Psi_{p0}.
\end{aligned} \tag{15}$$

因此，Ψ_{p0} 满足微分方程

$$\mathrm{i}\hbar\frac{\mathrm{d}}{\mathrm{d}p}\Psi_{p0} + \frac{\mathrm{i}}{m\omega}p\Psi_{p0} = 0, \tag{16}$$

分离变量得：

$$\frac{\mathrm{d}\Psi_{p0}}{\Psi_{p0}} = -\frac{1}{m\omega\hbar}p\mathrm{d}p. \tag{17}$$

两边积分得到通解

$$\Psi_{p0} = C_0\exp\left(-\frac{p^2}{2m\omega\hbar}\right). \tag{18}$$

利用高斯积分公式

$$\int_{-\infty}^{+\infty}\mathrm{e}^{-ax^2}\mathrm{d}x = \sqrt{\frac{\pi}{a}}, \tag{19}$$

利用归一化条件

$$\int_{-\infty}^{+\infty}|\Psi_{p0}|^2\,\mathrm{d}p = 1, \tag{20}$$

归一化系数可取为

$$C_0 = (\pi m\omega\hbar)^{-1/4}, \tag{21}$$

于是

$$\Psi_{p0} = (\pi m\omega\hbar)^{-1/4} \exp\left(-\frac{p^2}{2m\omega\hbar}\right). \tag{22}$$

产生算符同样可有坐标、动量算符表示：

$$\hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} - \frac{\mathrm{i}}{m\omega}\hat{p}\right). \tag{23}$$

于是

$$\begin{aligned}\Psi_{p1} = \langle p\,|\,1\rangle &= \left\langle p\,\middle|\,\hat{a}^\dagger\,\middle|\,0\right\rangle = \sqrt{\frac{m\omega}{2\hbar}}\left(\langle p\,|\,\hat{x}\,|\,0\rangle - \frac{\mathrm{i}}{m\omega}\langle p\,|\,\hat{p}\,|\,0\rangle\right)\\ &= \sqrt{\frac{m\omega}{2\hbar}}\left(\mathrm{i}\hbar\frac{\mathrm{d}}{\mathrm{d}p}\Psi_{p0} - \frac{\mathrm{i}}{m\omega}p\Psi_{p0}\right) = \sqrt{\frac{m\omega}{2\hbar}}\frac{(-2\mathrm{i})}{m\omega}p\Psi_{p0} = -\mathrm{i}\sqrt{\frac{2}{m\omega\hbar}}p\Psi_{p0}\\ &= -\mathrm{i}\sqrt{\frac{2}{m\omega\hbar}}\,(\pi m\omega\hbar)^{-1/4}\,p\exp\left(-\frac{p^2}{2m\omega\hbar}\right).\end{aligned} \tag{24}$$

题目八

(a) 求证谐振子系统坐标和动量算符本征态的下列展开式：

$$\begin{aligned}|x\rangle &= \sum_{n=0}^{\infty}|n\rangle\langle n\,|\,x\rangle\\ &= \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\exp\left[-\frac{m\omega}{2\hbar}x^2 + \sqrt{\frac{2m\omega}{\hbar}}x\hat{a}^\dagger - \frac{\left(\hat{a}^\dagger\right)^2}{2}\right]|0\rangle,\end{aligned} \tag{25}$$

$$\begin{aligned}|p\rangle &= \sum_{n=0}^{\infty}|n\rangle\langle n\,|\,p\rangle\\ &= \left(\frac{1}{m\omega\pi\hbar}\right)^{1/4}\exp\left[-\frac{p^2}{2m\omega\hbar} + \sqrt{\frac{2}{m\omega\hbar}}\,\mathrm{i}p\hat{a}^\dagger + \frac{\left(\hat{a}^\dagger\right)^2}{2}\right]|0\rangle.\end{aligned} \tag{26}$$

(b) 验证上述表达式满足

$$\hat{x}\,|x\rangle = x\,|x\rangle, \qquad \hat{p}\,|p\rangle = p\,|p\rangle. \tag{27}$$

回答

(a) 谐振子本征态坐标表象波函数为

$$\psi_n(x) = \langle x\,|\,n\rangle = \frac{1}{\pi^{1/4}\sqrt{x_0}}\frac{H_n(\xi)}{\sqrt{2^n n!}}\mathrm{e}^{-\xi^2/2}, \quad x_0 = \sqrt{\frac{\hbar}{m\omega}}, \quad \xi = \frac{x}{x_0} = \sqrt{\frac{m\omega}{\hbar}}x. \tag{28}$$

利用厄米多项式的生成函数公式

$$\mathrm{e}^{-t^2+2xt} = \sum_{n=0}^{\infty}\frac{H_n(x)}{n!}t^n, \tag{29}$$

可得

$$\begin{aligned}
|x\rangle &= \sum_{n=0}^{\infty} \langle n \,|\, x\rangle \,|n\rangle = \sum_{n=0}^{\infty} \psi_n^*(x) \frac{\left(\hat{a}^\dagger\right)^n}{\sqrt{n!}} |0\rangle = \frac{1}{\pi^{1/4}\sqrt{x_0}} \mathrm{e}^{-\xi^2/2} \left[\sum_{n=0}^{\infty} \frac{H_n(\xi)}{n!} \left(\frac{\hat{a}^\dagger}{\sqrt{2}}\right)^n\right] |0\rangle \\
&= \frac{1}{\pi^{1/4}\sqrt{x_0}} \mathrm{e}^{-\xi^2/2} \exp\left[-\frac{\left(\hat{a}^\dagger\right)^2}{2} + \sqrt{2}\xi\hat{a}^\dagger\right] |0\rangle \\
&= \frac{1}{\pi^{1/4}\sqrt{x_0}} \exp\left[-\frac{\left(\hat{a}^\dagger\right)^2}{2} + \sqrt{2}\xi\hat{a}^\dagger - \frac{\xi^2}{2}\right] |0\rangle \\
&= \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \exp\left[-\frac{m\omega}{2\hbar}x^2 + \sqrt{\frac{2m\omega}{\hbar}}x\hat{a}^\dagger - \frac{\left(\hat{a}^\dagger\right)^2}{2}\right] |0\rangle .
\end{aligned} \tag{30}$$

利用高斯积分公式

$$\int_{-\infty}^{+\infty} \mathrm{e}^{-Ax^2+Bx} \mathrm{d}x = \sqrt{\frac{\pi}{A}} \exp\left(\frac{B^2}{4A}\right), \tag{31}$$

可得

$$\begin{aligned}
|p\rangle &= \int \mathrm{d}x \,\langle x \,|\, p\rangle \,|x\rangle \\
&= \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \frac{1}{\sqrt{2\pi\hbar}} \int \mathrm{d}x \mathrm{e}^{\mathrm{i}px/\hbar} \exp\left[-\frac{m\omega}{2\hbar}x^2 + \sqrt{\frac{2m\omega}{\hbar}}x\hat{a}^\dagger - \frac{\left(\hat{a}^\dagger\right)^2}{2}\right] |0\rangle \\
&= \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \frac{1}{\sqrt{2\pi\hbar}} \int \mathrm{d}x \exp\left[-\frac{m\omega}{2\hbar}x^2 + \left(\sqrt{\frac{2m\omega}{\hbar}}\hat{a}^\dagger + \frac{\mathrm{i}p}{\hbar}\right)x - \frac{\left(\hat{a}^\dagger\right)^2}{2}\right] |0\rangle \\
&= \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \frac{1}{\sqrt{2\pi\hbar}} \sqrt{\frac{2\pi\hbar}{m\omega}} \exp\left\{\frac{\hbar}{2m\omega}\left[\frac{2m\omega}{\hbar}\left(\hat{a}^\dagger\right)^2 + \frac{2\mathrm{i}}{\hbar}\sqrt{\frac{2m\omega}{\hbar}}p\hat{a}^\dagger - \frac{p^2}{\hbar^2}\right]\right\} \exp\left[-\frac{\left(\hat{a}^\dagger\right)^2}{2}\right] |0\rangle \\
&= \left(\frac{1}{m\omega\pi\hbar}\right)^{1/4} \exp\left[-\frac{p^2}{2m\omega\hbar} + \sqrt{\frac{2}{m\omega\hbar}}\,\mathrm{i}p\hat{a}^\dagger + \frac{\left(\hat{a}^\dagger\right)^2}{2}\right] |0\rangle .
\end{aligned} \tag{32}$$

(b) 由

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} + \frac{\mathrm{i}}{m\omega}\hat{p}\right), \qquad \hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} - \frac{\mathrm{i}}{m\omega}\hat{p}\right), \tag{33}$$

可以反解出

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}\left(\hat{a} + \hat{a}^\dagger\right), \qquad \hat{p} = -\mathrm{i}\sqrt{\frac{\hbar m\omega}{2}}\left(\hat{a} - \hat{a}^\dagger\right). \tag{34}$$

利用

$$\hat{a} f\left(\hat{a}^\dagger\right) |0\rangle = \left[\hat{a}, f\left(\hat{a}^\dagger\right)\right] |0\rangle , \qquad \left[\hat{a}, f\left(\hat{a}^\dagger\right)\right] = f'\left(\hat{a}^\dagger\right), \tag{35}$$

令

$$F\left(\hat{a}^\dagger\right) = \exp\left[-\frac{m\omega}{2\hbar}x^2 + \sqrt{\frac{2m\omega}{\hbar}}x\hat{a}^\dagger - \frac{\left(\hat{a}^\dagger\right)^2}{2}\right], \tag{36}$$

$$\begin{aligned}
F'\left(\hat{a}^\dagger\right) &= \left(\sqrt{\frac{2m\omega}{\hbar}}x - \hat{a}^\dagger\right) \exp\left[-\frac{m\omega}{2\hbar}x^2 + \sqrt{\frac{2m\omega}{\hbar}}x\hat{a}^\dagger - \frac{\left(\hat{a}^\dagger\right)^2}{2}\right] \\
&= \left(\sqrt{\frac{2m\omega}{\hbar}}x - \hat{a}^\dagger\right) F\left(\hat{a}^\dagger\right)
\end{aligned} \tag{37}$$

可得

$$\begin{aligned}
\hat{x}\left|x\right\rangle &= \sqrt{\frac{\hbar}{2m\omega}}\left(\hat{a}+\hat{a}^{\dagger}\right)\left\{\left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\exp\left[-\frac{m\omega}{2\hbar}x^{2}+\sqrt{\frac{2m\omega}{\hbar}}x\hat{a}^{\dagger}-\frac{\left(\hat{a}^{\dagger}\right)^{2}}{2}\right]\left|0\right\rangle\right\} \\
&= \sqrt{\frac{\hbar}{2m\omega}}\left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\left(\hat{a}+\hat{a}^{\dagger}\right)F\left(\hat{a}^{\dagger}\right)\left|0\right\rangle \\
&= \sqrt{\frac{\hbar}{2m\omega}}\left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\left\{\left[\hat{a},F\left(\hat{a}^{\dagger}\right)\right]+\hat{a}^{\dagger}F\left(\hat{a}^{\dagger}\right)\right\}\left|0\right\rangle \\
&= \sqrt{\frac{\hbar}{2m\omega}}\left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\left\{F'\left(\hat{a}^{\dagger}\right)+\hat{a}^{\dagger}F\left(\hat{a}^{\dagger}\right)\right\}\left|0\right\rangle \\
&= \sqrt{\frac{\hbar}{2m\omega}}\left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\left\{\left(\sqrt{\frac{2m\omega}{\hbar}}x-\hat{a}^{\dagger}\right)F\left(\hat{a}^{\dagger}\right)+\hat{a}^{\dagger}F\left(\hat{a}^{\dagger}\right)\right\}\left|0\right\rangle \\
&= x\left(\frac{m\omega}{\pi\hbar}\right)^{1/4}F\left(\hat{a}^{\dagger}\right)\left|0\right\rangle \\
&= x\left|x\right\rangle .
\end{aligned} \tag{38}$$

令

$$G\left(\hat{a}^{\dagger}\right)=\exp\left[-\frac{p^{2}}{2m\omega\hbar}+\sqrt{\frac{2}{m\omega\hbar}}\,\mathrm{i}p\hat{a}^{\dagger}+\frac{\left(\hat{a}^{\dagger}\right)^{2}}{2}\right] \tag{39}$$

$$G'\left(\hat{a}^{\dagger}\right)=\left(\sqrt{\frac{2}{m\omega\hbar}}\,\mathrm{i}p+\hat{a}^{\dagger}\right)G\left(\hat{a}^{\dagger}\right), \tag{40}$$

可得

$$\begin{aligned}
\hat{p}\left|p\right\rangle &= -\mathrm{i}\sqrt{\frac{\hbar m\omega}{2}}\left(\hat{a}-\hat{a}^{\dagger}\right)\left\{\left(\frac{1}{m\omega\pi\hbar}\right)^{1/4}\exp\left[-\frac{p^{2}}{2m\omega\hbar}+\sqrt{\frac{2}{m\omega\hbar}}\,\mathrm{i}p\hat{a}^{\dagger}+\frac{\left(\hat{a}^{\dagger}\right)^{2}}{2}\right]\left|0\right\rangle\right\} \\
&= -\mathrm{i}\sqrt{\frac{\hbar m\omega}{2}}\left(\frac{1}{m\omega\pi\hbar}\right)^{1/4}\left(\hat{a}-\hat{a}^{\dagger}\right)G\left(\hat{a}^{\dagger}\right)\left|0\right\rangle \\
&= -\mathrm{i}\sqrt{\frac{\hbar m\omega}{2}}\left(\frac{1}{m\omega\pi\hbar}\right)^{1/4}\left\{\left[\hat{a},G\left(\hat{a}^{\dagger}\right)\right]-\hat{a}^{\dagger}G\left(\hat{a}^{\dagger}\right)\right\}\left|0\right\rangle \\
&= -\mathrm{i}\sqrt{\frac{\hbar m\omega}{2}}\left(\frac{1}{m\omega\pi\hbar}\right)^{1/4}\left\{G'\left(\hat{a}^{\dagger}\right)-\hat{a}^{\dagger}G\left(\hat{a}^{\dagger}\right)\right\}\left|0\right\rangle \\
&= -\mathrm{i}\sqrt{\frac{\hbar m\omega}{2}}\left(\frac{1}{m\omega\pi\hbar}\right)^{1/4}\left\{\left(\sqrt{\frac{2}{m\omega\hbar}}\,\mathrm{i}p+\hat{a}^{\dagger}\right)G\left(\hat{a}^{\dagger}\right)-\hat{a}^{\dagger}G\left(\hat{a}^{\dagger}\right)\right\}\left|0\right\rangle \\
&= p\left(\frac{1}{m\omega\pi\hbar}\right)^{1/4}G\left(\hat{a}^{\dagger}\right)\left|0\right\rangle \\
&= p\left|p\right\rangle .
\end{aligned} \tag{41}$$

题目九 位移算符 $\hat{D}\left(\alpha\right)$ 定义为

$$\hat{D}\left(\alpha\right)=\exp\left(\alpha\hat{a}^{\dagger}-\alpha^{*}\hat{a}\right). \tag{42}$$

(a) 证明

$$\hat{D}\left(\alpha\right)\hat{a}\hat{D}^{\dagger}\left(\alpha\right)=\hat{a}-\alpha,\qquad\hat{D}\left(\alpha\right)\hat{a}^{\dagger}\hat{D}^{\dagger}\left(\alpha\right)=\hat{a}^{\dagger}-\alpha^{*}. \tag{43}$$

(b) 求

$$\hat{D}(\alpha)\hat{D}(\beta)\hat{x}^2\hat{D}^\dagger(\beta)\hat{D}^\dagger(\alpha). \tag{44}$$

回答

(a) Hadamard 引理给出：

$$\mathrm{e}^A B \mathrm{e}^{-A} = B + [A,B] + \frac{1}{2!}[A,[A,B]] + \cdots. \tag{45}$$

令

$$A = \alpha a^\dagger - \alpha^* a, \qquad D(\alpha) = \mathrm{e}^A, \qquad D^\dagger(\alpha) = \mathrm{e}^{-A}. \tag{46}$$

注意到

$$[A,a] = -\alpha, \qquad [A,[A,a]] = 0, \qquad \cdots, \tag{47}$$

于是有

$$D(\alpha) a D^\dagger(\alpha) = \mathrm{e}^A a \mathrm{e}^{-A} = a - \alpha. \tag{48}$$

注意到

$$\left[A, a^\dagger\right] = -\alpha^*, \qquad \left[A, \left[A, a^\dagger\right]\right] = 0, \qquad \cdots, \tag{49}$$

于是有

$$D(\alpha) a^\dagger D^\dagger(\alpha) = \mathrm{e}^A a^\dagger \mathrm{e}^{-A} = a^\dagger - \alpha^*. \tag{50}$$

(b) 由

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}\left(\hat{a} + \hat{a}^\dagger\right) \tag{51}$$

可得

$$\begin{aligned} \hat{D}(\alpha)\hat{x}\hat{D}^\dagger(\alpha) &= \sqrt{\frac{\hbar}{2m\omega}}\hat{D}(\alpha)\left(\hat{a} + \hat{a}^\dagger\right)\hat{D}^\dagger(\alpha) = \sqrt{\frac{\hbar}{2m\omega}}\left(\hat{a} + \hat{a}^\dagger - \alpha - \alpha^*\right) \\ &= \hat{x} - \sqrt{\frac{\hbar}{2m\omega}}(\alpha + \alpha^*) \equiv \hat{x} - x_\alpha. \end{aligned} \tag{52}$$

于是

$$\begin{aligned} \hat{D}(\alpha)\hat{D}(\beta)\hat{x}^2\hat{D}^\dagger(\beta)\hat{D}^\dagger(\alpha) &= \hat{D}(\alpha)\hat{D}(\beta)\hat{x}\hat{D}^\dagger(\beta)\hat{D}(\beta)\hat{x}\hat{D}^\dagger(\beta)\hat{D}^\dagger(\alpha) \\ &= \hat{D}(\alpha)(\hat{x} - x_\beta)(\hat{x} - x_\beta)\hat{D}^\dagger(\alpha) \\ &= \hat{D}(\alpha)(\hat{x} - x_\beta)\hat{D}^\dagger(\alpha)\hat{D}(\alpha)(\hat{x} - x_\beta)\hat{D}^\dagger(\alpha) \\ &= (\hat{x} - x_\alpha - x_\beta)^2 \\ &= \left[\hat{x} - \sqrt{\frac{\hbar}{2m\omega}}(\alpha + \alpha^* + \beta + \beta^*)\right]^2. \end{aligned} \tag{53}$$

题目十 求在相干态表象下，算符 $\hat{x}$ 和 $\hat{p}$ 的外积表达式。

回答 相干态是湮灭算符本征态：

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle, \qquad \langle\alpha|\hat{a}^\dagger = \alpha^*\langle\alpha|. \tag{54}$$

由相干态完备性关系

$$\int \frac{\mathrm{d}^2\alpha}{\pi} |\alpha\rangle \langle\alpha| = \hat{I}, \tag{55}$$

可得

$$\hat{a} = \hat{a}\hat{I} = \int \frac{\mathrm{d}^2\alpha}{\pi} \hat{a} |\alpha\rangle \langle\alpha| = \int \frac{\mathrm{d}^2\alpha}{\pi} \alpha |\alpha\rangle \langle\alpha| , \tag{56}$$

$$\hat{a}^\dagger = \hat{I}\hat{a}^\dagger = \int \frac{\mathrm{d}^2\alpha}{\pi} |\alpha\rangle \langle\alpha| \hat{a}^\dagger = \int \frac{\mathrm{d}^2\alpha}{\pi} \alpha^* |\alpha\rangle \langle\alpha| . \tag{57}$$

因此

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}} \left(\hat{a} + \hat{a}^\dagger\right) = \sqrt{\frac{\hbar}{2m\omega}} \int \frac{\mathrm{d}^2\alpha}{\pi} (\alpha + \alpha^*) |\alpha\rangle \langle\alpha| , \tag{58}$$

$$\hat{p} = -\mathrm{i}\sqrt{\frac{\hbar m\omega}{2}} \left(\hat{a} - \hat{a}^\dagger\right) = -\mathrm{i}\sqrt{\frac{\hbar m\omega}{2}} \int \frac{\mathrm{d}^2\alpha}{\pi} (\alpha - \alpha^*) |\alpha\rangle \langle\alpha| . \tag{59}$$

题目十一 计算在相干态下，谐振子系统粒子动量的几率分布 $|\langle p | \alpha\rangle|^2$。

回答 考虑相干态本征方程：

$$\hat{a} |\alpha\rangle = \alpha |\alpha\rangle . \tag{60}$$

记

$$\langle p | \alpha\rangle \equiv \psi_\alpha(p), \tag{61}$$

本征方程两边左乘 $\langle p|$ 得

$$\langle p | \hat{a} | \alpha\rangle = \alpha \langle p | \alpha\rangle = \alpha\psi_\alpha(p). \tag{62}$$

利用

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} + \frac{\mathrm{i}}{m\omega}\hat{p}\right), \tag{63}$$

以及

$$\langle p | \hat{x} | \alpha\rangle = \mathrm{i}\hbar \frac{\partial}{\partial p} \langle p | \alpha\rangle = \mathrm{i}\hbar \frac{\mathrm{d}}{\mathrm{d}p}\psi_\alpha(p), \tag{64}$$

$$\langle p | \hat{p} | \alpha\rangle = p \langle p | \alpha\rangle = p\psi_\alpha(p), \tag{65}$$

代入(62)可得：

$$\sqrt{\frac{m\omega}{2\hbar}} \left[\mathrm{i}\hbar \frac{\mathrm{d}}{\mathrm{d}p}\psi_\alpha(p) + \frac{\mathrm{i}}{m\omega} p\psi_\alpha(p)\right] = \alpha\psi_\alpha(p). \tag{66}$$

也即：

$$\frac{\mathrm{d}\psi_\alpha(p)}{\psi_\alpha(p)} = \left[-\frac{1}{\hbar m\omega}p - \mathrm{i}\sqrt{\frac{2}{\hbar m\omega}}\alpha\right] \mathrm{d}p, \tag{67}$$

两边积分得到通解：

$$\psi_\alpha(p) = C \exp\left(-\frac{1}{2\hbar m\omega}p^2 - \mathrm{i}\sqrt{\frac{2}{\hbar m\omega}}\alpha p\right). \tag{68}$$

设

$$\alpha = \alpha_\mathrm{R} + \mathrm{i}\alpha_\mathrm{I}, \qquad \alpha_\mathrm{R}, \alpha_\mathrm{I} \in \mathbb{R}, \tag{69}$$

则

$$|\psi_\alpha(p)|^2 = |C|^2 \exp\left(-\frac{1}{\hbar m\omega}p^2 + 2\sqrt{\frac{2}{\hbar m\omega}}\alpha_\mathrm{I} p\right), \tag{70}$$

由归一化条件

$$\int_{-\infty}^{+\infty} |\psi_\alpha(p)|^2 \,\mathrm{d}p = 1, \tag{71}$$

以及高斯积分公式

$$\int_{-\infty}^{+\infty} \mathrm{e}^{-Ax^2+Bx}\mathrm{d}x = \sqrt{\frac{\pi}{A}}\exp\left(\frac{B^2}{4A}\right), \qquad A>0, \tag{72}$$

可得归一化系数满足

$$|C|^2 = (\pi\hbar m\omega)^{-1/2}\exp\left(-2\alpha_{\mathrm{I}}^2\right). \tag{73}$$

因此，在相干态下，谐振子系统粒子动量的几率分布为：

$$\begin{aligned} |\langle p\,|\,\alpha\rangle|^2 = |\psi_\alpha(p)|^2 &= \frac{1}{\sqrt{\pi\hbar m\omega}}\exp\left(-\frac{1}{\hbar m\omega}p^2 + 2\sqrt{\frac{2}{\hbar m\omega}}\alpha_{\mathrm{I}}p - 2\alpha_{\mathrm{I}}^2\right) \\ &= \frac{1}{\sqrt{\pi\hbar m\omega}}\exp\left[-\left(\frac{1}{\sqrt{\hbar m\omega}}p - \sqrt{2}\mathrm{Im}\,\alpha\right)^2\right]. \end{aligned} \tag{74}$$

题目十二 写出关于谐振子第一激发态 $|1\rangle$ 的任意三种不同的相干态展开式。

回答 由于

$$\hat{a}\,|n\rangle = \sqrt{n}\,|n-1\rangle, \qquad \hat{a}^\dagger\,|n\rangle = \sqrt{n+1}\,|n+1\rangle, \tag{75}$$

于是

$$\hat{a}^k\left(\hat{a}^\dagger\right)^k|1\rangle = \sqrt{2}\sqrt{3}\cdots\sqrt{k+1}\hat{a}^k\,|k+1\rangle = \left(\sqrt{2}\sqrt{3}\cdots\sqrt{k+1}\right)^2|1\rangle = (k+1)!\,|1\rangle, \tag{76}$$

$$|1\rangle = \frac{1}{(k+1)!}\hat{a}^k\left(\hat{a}^\dagger\right)^k|1\rangle. \tag{77}$$

在湮灭、产生算符中间插入相干态完备性关系：

$$\begin{aligned} |1\rangle &= \frac{1}{(k+1)!}\int\frac{\mathrm{d}^2\alpha}{\pi}\hat{a}^k\,|\alpha\rangle\langle\alpha|\left(\hat{a}^\dagger\right)^k|1\rangle \\ &= \frac{1}{(k+1)!}\int\frac{\mathrm{d}^2\alpha}{\pi}\alpha^k\,|\alpha\rangle\langle\alpha|\,(\alpha^*)^k\,|1\rangle \\ &= \frac{1}{(k+1)!}\int\frac{\mathrm{d}^2\alpha}{\pi}|\alpha|^{2k}\,|\alpha\rangle\langle\alpha\,|\,1\rangle \end{aligned} \tag{78}$$

把相干态用数态展开：

$$|\alpha\rangle = \mathrm{e}^{-|\alpha|^2/2}\sum_{n=0}^{\infty}\frac{\alpha^n}{\sqrt{n!}}\,|n\rangle, \tag{79}$$

两边左乘 $\langle 1|$ 得：

$$\langle 1\,|\,\alpha\rangle = \alpha\mathrm{e}^{-|\alpha|^2/2}, \qquad \langle\alpha\,|\,1\rangle = \langle 1\,|\,\alpha\rangle^* = \alpha^*\mathrm{e}^{-|\alpha|^2/2}, \tag{80}$$

于是

$$\begin{aligned} |1\rangle &= \frac{1}{(k+1)!}\int\frac{\mathrm{d}^2\alpha}{\pi}|\alpha|^{2k}\,|\alpha\rangle\langle\alpha\,|\,1\rangle \\ &= \frac{1}{(k+1)!}\int\frac{\mathrm{d}^2\alpha}{\pi}|\alpha|^{2k}\,\mathrm{e}^{-|\alpha|^2/2}\alpha^*\,|\alpha\rangle. \end{aligned} \tag{81}$$

分别取 $k=0,1,2$ 就得到 $|1\rangle$ 在相干态表象下的三种展开式：

$$|1\rangle = \int \frac{\mathrm{d}^2\alpha}{\pi} \mathrm{e}^{-|\alpha|^2/2} \alpha^* |\alpha\rangle \,, \tag{82}$$

$$|1\rangle = \frac{1}{2} \int \frac{\mathrm{d}^2\alpha}{\pi} |\alpha|^2 \, \mathrm{e}^{-|\alpha|^2/2} \alpha^* |\alpha\rangle \,, \tag{83}$$

$$|1\rangle = \frac{1}{6} \int \frac{\mathrm{d}^2\alpha}{\pi} |\alpha|^4 \, \mathrm{e}^{-|\alpha|^2/2} \alpha^* |\alpha\rangle \,. \tag{84}$$

题目十三 证明

$$\int_{\mathbb{C}} \frac{\mathrm{d}^2\alpha}{\pi} \mathrm{e}^{\beta^*\alpha - |\alpha|^2} (\alpha^*)^n = (\beta^*)^n \,. \tag{85}$$

回答 令

$$I_n = \int \frac{\mathrm{d}^2\alpha}{\pi} \mathrm{e}^{\beta^*\alpha - |\alpha|^2} (\alpha^*)^n \,, \tag{86}$$

考虑 $\mathrm{e}^{\beta^*\alpha}$ 的级数展开，有

$$\begin{aligned} I_n &= \int \frac{\mathrm{d}^2\alpha}{\pi} \mathrm{e}^{\beta^*\alpha - |\alpha|^2} (\alpha^*)^n = \int \frac{\mathrm{d}^2\alpha}{\pi} \left(\sum_{m=0}^{\infty} \frac{(\beta^*)^m \alpha^m}{m!} \right) \mathrm{e}^{-|\alpha|^2} (\alpha^*)^n \\ &= \sum_{m=0}^{\infty} \frac{(\beta^*)^m}{m!} \int \frac{\mathrm{d}^2\alpha}{\pi} \mathrm{e}^{-|\alpha|^2} \alpha^m (\alpha^*)^n \,. \end{aligned} \tag{87}$$

令

$$J_{mn} = \int \frac{\mathrm{d}^2\alpha}{\pi} \mathrm{e}^{-|\alpha|^2} \alpha^m (\alpha^*)^n \,, \tag{88}$$

考虑极坐标

$$\alpha = r\mathrm{e}^{\mathrm{i}\phi}, \qquad \mathrm{d}^2\alpha = r\mathrm{d}r\mathrm{d}\phi, \tag{89}$$

则

$$\begin{aligned} J_{mn} &= \int \frac{\mathrm{d}^2\alpha}{\pi} \mathrm{e}^{-|\alpha|^2} \alpha^m (\alpha^*)^n = \int \frac{r\mathrm{d}r\mathrm{d}\phi}{\pi} \mathrm{e}^{-r^2} \left(r\mathrm{e}^{\mathrm{i}\phi}\right)^m \left(r\mathrm{e}^{-\mathrm{i}\phi}\right)^n \\ &= \frac{1}{\pi} \int_0^{\infty} \mathrm{d}r \, r^{m+n+1} \mathrm{e}^{-r^2} \int_0^{2\pi} \mathrm{d}\phi \, \mathrm{e}^{\mathrm{i}(m-n)\phi} = \frac{1}{\pi} \int_0^{\infty} \mathrm{d}r \, r^{m+n+1} \mathrm{e}^{-r^2} \cdot 2\pi\delta_{mn} \\ &= 2\delta_{mn} \int_0^{\infty} \mathrm{d}r \, r^{2n+1} \mathrm{e}^{-r^2}. \end{aligned} \tag{90}$$

令

$$u = r^2, \qquad \mathrm{d}u = 2r\mathrm{d}r, \tag{91}$$

则

$$J_{mn} = 2\delta_{mn} \int_0^{\infty} \mathrm{d}r \, r^{2n+1} \mathrm{e}^{-r^2} = \delta_{mn} \int_0^{\infty} \mathrm{d}u \, u^n \mathrm{e}^{-u} = \delta_{mn} \Gamma(n+1) = \delta_{mn} n!. \tag{92}$$

于是

$$\begin{aligned} I_n &= \sum_{m=0}^{\infty} \frac{(\beta^*)^m}{m!} \int \frac{\mathrm{d}^2\alpha}{\pi} \mathrm{e}^{-|\alpha|^2} \alpha^m (\alpha^*)^n = \sum_{m=0}^{\infty} \frac{(\beta^*)^m}{m!} J_{mn} \\ &= \sum_{m=0}^{\infty} \frac{(\beta^*)^m}{m!} \delta_{mn} n! = (\beta^*)^n \,. \end{aligned} \tag{93}$$

题目十四 通过计算表达式的右方，验证谐振子粒子数算符的本征态 $|n\rangle$ 的如下展开式：

(a)

$$|n\rangle = \int_{\mathbb{C}} \frac{\mathrm{d}^2\alpha}{\pi} |\alpha\rangle \exp\left(-|\alpha|^2/2\right) \frac{(\alpha^*)^n}{\sqrt{n!}}. \tag{94}$$

(b)

$$|n\rangle = \frac{1}{2\pi} \exp\left(\frac{r_0^2}{2}\right) r_0^{-n}\sqrt{n!} \int_0^{2\pi} \mathrm{e}^{-\mathrm{i}n\phi} \left|r_0\mathrm{e}^{\mathrm{i}\phi}\right\rangle \mathrm{d}\phi, \tag{95}$$

其中 $\alpha \equiv r_0\mathrm{e}^{\mathrm{i}\phi}$。

回答

(a) 把相干态用数态展开，再利用 (92)，有

$$\begin{aligned}
&\int \frac{\mathrm{d}^2\alpha}{\pi} |\alpha\rangle \exp\left(-|\alpha|^2/2\right) \frac{(\alpha^*)^n}{\sqrt{n!}} \\
=&\int \frac{\mathrm{d}^2\alpha}{\pi} \left(\mathrm{e}^{-|\alpha|^2/2} \sum_{m=0}^{\infty} \frac{\alpha^m}{\sqrt{m!}} |m\rangle\right) \exp\left(-|\alpha|^2/2\right) \frac{(\alpha^*)^n}{\sqrt{n!}} \\
=&\sum_{m=0}^{\infty} \frac{1}{\sqrt{m!n!}} \int \frac{\mathrm{d}^2\alpha}{\pi} \mathrm{e}^{-|\alpha|^2} \alpha^m (\alpha^*)^n |m\rangle = \sum_{m=0}^{\infty} \frac{1}{\sqrt{m!n!}} J_{mn} |m\rangle \\
=&\sum_{m=0}^{\infty} \frac{1}{\sqrt{m!n!}} (\delta_{mn} n!) |m\rangle \\
=&|n\rangle .
\end{aligned} \tag{96}$$

(b) 相干态按数态展开：

$$|\alpha\rangle = \left|r_0\mathrm{e}^{\mathrm{i}\phi}\right\rangle = \left(\mathrm{e}^{-|\alpha|^2/2} \sum_{m=0}^{\infty} \frac{\alpha^m}{\sqrt{m!}} |m\rangle\right) = \mathrm{e}^{-r_0^2/2} \sum_{m=0}^{\infty} \frac{r_0^m \mathrm{e}^{\mathrm{i}m\phi}}{\sqrt{m!}} |m\rangle , \tag{97}$$

因此

$$\begin{aligned}
&\frac{1}{2\pi} \exp\left(\frac{r_0^2}{2}\right) r_0^{-n}\sqrt{n!} \int_0^{2\pi} \mathrm{e}^{-\mathrm{i}n\phi} \left|r_0\mathrm{e}^{\mathrm{i}\phi}\right\rangle \mathrm{d}\phi \\
=&\frac{1}{2\pi} \exp\left(\frac{r_0^2}{2}\right) r_0^{-n}\sqrt{n!} \int_0^{2\pi} \mathrm{e}^{-\mathrm{i}n\phi} \left(\mathrm{e}^{-r_0^2/2} \sum_{m=0}^{\infty} \frac{r_0^m \mathrm{e}^{\mathrm{i}m\phi}}{\sqrt{m!}} |m\rangle\right) \mathrm{d}\phi \\
=&\frac{1}{2\pi} r_0^{-n}\sqrt{n!} \sum_{m=0}^{\infty} \frac{r_0^m}{\sqrt{m!}} |m\rangle \int_0^{2\pi} \mathrm{e}^{\mathrm{i}(m-n)\phi} \mathrm{d}\phi \\
=&\frac{1}{2\pi} r_0^{-n}\sqrt{n!} \sum_{m=0}^{\infty} \frac{r_0^m}{\sqrt{m!}} |m\rangle \cdot 2\pi\delta_{mn} \\
=&|n\rangle .
\end{aligned} \tag{98}$$

题目十五 设粒子在 δ 势阱中运动，Hamilton 算符为

$$\hat{H} = \frac{\hat{p}^2}{2m} - g\delta(\hat{x}) . \tag{99}$$

试在动量表象中求解束缚能级 $E<0$ 和对应本征态。

回答 设

$$\hat{H}\left|\psi\right\rangle=\left[\frac{\hat{p}^2}{2m}-g\delta\left(\hat{x}\right)\right]\left|\psi\right\rangle=E\left|\psi\right\rangle, \tag{100}$$

两边左乘 $\langle p|$ 得：

$$\frac{p^2}{2m}\left\langle p\left|\psi\right.\right\rangle-g\left\langle p\left|\delta\left(\hat{x}\right)\right|\psi\right\rangle=E\left\langle p\left|\psi\right.\right\rangle. \tag{101}$$

记

$$\psi(p)=\left\langle p\left|\psi\right.\right\rangle, \tag{102}$$

注意到

$$\begin{aligned}\left\langle p\left|\delta\left(\hat{x}\right)\right|\psi\right\rangle&=\int\mathrm{d}p'\left\langle p\left|\delta\left(\hat{x}\right)\right|p'\right\rangle\left\langle p'\left|\psi\right.\right\rangle=\int\mathrm{d}p'\int\mathrm{d}x\left\langle p\left|\delta\left(\hat{x}\right)\right|x\right\rangle\left\langle x\left|p'\right.\right\rangle\psi(p')\\&=\int\mathrm{d}p'\int\mathrm{d}x\delta\left(x\right)\left\langle p\left|x\right.\right\rangle\left\langle x\left|p'\right.\right\rangle\psi(p')\\&=\int\mathrm{d}p'\left\langle p\left|x=0\right.\right\rangle\left\langle x=0\left|p'\right.\right\rangle\psi(p')\\&=\frac{1}{2\pi\hbar}\int_{-\infty}^{+\infty}\mathrm{d}p'\psi(p')\equiv\frac{I}{2\pi\hbar},\qquad I=\int_{-\infty}^{+\infty}\mathrm{d}p'\psi(p')\end{aligned} \tag{103}$$

因此有

$$\psi(p)=\frac{gI}{2\pi\hbar\left(\frac{p^2}{2m}-E\right)}, \tag{104}$$

两边对 p 积分：

$$I=\frac{gI}{2\pi\hbar}\int_{-\infty}^{+\infty}\frac{1}{\frac{p^2}{2m}-E}\mathrm{d}p=\frac{gI}{2\pi\hbar}\cdot\pi\sqrt{\frac{2m}{-E}}, \tag{105}$$

对比头尾可得本征能量：

$$E=-\frac{mg^2}{2\hbar^2} \tag{106}$$

设动量表象本征波函数归一化为

$$\psi(p)=\frac{A}{\left(\frac{p^2}{2m}-E\right)}, \tag{107}$$

由归一化条件

$$\int_{-\infty}^{+\infty}\left|\psi(p)\right|^2\mathrm{d}p=1 \tag{108}$$

可得 A 可取为

$$A=\sqrt{\frac{m}{2\pi}}\left(\frac{g}{\hbar}\right)^{3/2}, \tag{109}$$

于是

$$\psi(p)=\sqrt{\frac{m}{2\pi}}\left(\frac{g}{\hbar}\right)^{3/2}\frac{1}{\frac{p^2}{2m}+\frac{mg^2}{2\hbar^2}}. \tag{110}$$

态矢量为

$$\left|\psi\right\rangle=\int\mathrm{d}p\psi(p)\left|p\right\rangle=\sqrt{\frac{m}{2\pi}}\left(\frac{g}{\hbar}\right)^{3/2}\int\mathrm{d}p\frac{1}{\frac{p^2}{2m}+\frac{mg^2}{2\hbar^2}}\left|p\right\rangle=\sqrt{\frac{2}{\pi}}\left(\frac{mg}{\hbar}\right)^{3/2}\int\mathrm{d}p\frac{\left|p\right\rangle}{p^2+\left(gm/\hbar\right)^2}. \tag{111}$$